

CHRISTOPHER LINDSAY

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EDUCATION

Yale University <i>Astronomy PhD (M.S., M.Phil. in Astronomy, 2022)</i>	August 2020 - May 2026 New Haven, Connecticut, USA
University of Southern California (USC) <i>B.S. Astronomy; B.A. Environmental Studies; Jazz Studies</i>	August 2016 - May 2020 Los Angeles, California, USA

RESEARCH APPOINTMENTS

Postdoctoral Research (NSF AAPF Fellow) <i>Scientific Mentor: Jamie Tayar</i>	September 2026 - present The University of Florida
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- My postdoctoral research focuses on making precise asteroseismic measurements of stars that were accreted by the Milky Way to enable asteroseismic stellar age determinations for a large sample of halo stars. The ages will be used to constrain timescales for merger events in the Milky Way's history and study heavy element enhancement in dwarf galaxies.

Graduate Research (Gruber Science Fellow) <i>Advisor: Sarbani Basu</i>	August 2020 - May 2026 Yale University
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- My thesis research involved the analysis of stellar oscillation data from NASA's *Kepler* and TESS missions to model stellar interiors and determine fundamental stellar parameters with the goal of improving our understanding of the Milky Way's formation through asteroseismic age determinations.
- I have also advised and currently advise multiple Yale University undergraduate and more junior graduate students on stellar astrophysics projects related to machine learning detection of stellar/solar flares, RR Lyrae stellar modeling, population III star asteroseismology, and more.

Undergraduate Research <i>Advisor: Edward Rhodes</i>	August 2016 - August 2020 The University of Southern California
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- I collected heliosismic data at the Mount Wilson Observatory and processed solar oscillation data from the Solar Dynamics Observatory and Solar and Heliospheric Observatory satellites.

PUBLICATIONS

[[Listed Here in a Public ADS Library](#)]

First Author Papers

5. **Christopher Lindsay**, et al., 2026 (*ApJ*, [1000:232](#)) *The Effect of Different Methods for Accounting for α -enhancement on the Asteroseismic Modeling of Metal-Poor Stars.*
4. **Christopher Lindsay**, Marc Hon, et al., 2025 (*ApJ*, [989:189](#)) *Precise Asteroseismic Ages for the Helmi Streams*

3. **Christopher Lindsay**, Joel Ong, and Sarbani Basu, 2024 (*ApJ*, [965:171](#)) *Fossil Signatures of Main-sequence Convective Core Overshoot Estimated through Asteroseismic Analyses*
2. **Christopher Lindsay**, Joel Ong, and Sarbani Basu, 2023 (*ApJ*, [950:19](#)) *Near-Core Acoustic Glitches are Not Oscillatory: Consequences for Asteroseismic Probes of Convective Boundary Mixing*
1. **Christopher Lindsay**, Joel Ong, and Sarbani Basu, 2022 (*ApJ*, [931:116](#)) *Mixed Mode Asteroseismology of Red Giant Stars Through the Luminosity Bump*

Co-author Papers

10. R. A. García, S. Mathur, et al. (incl. **C. Lindsay**), 2026 (*in press, A&A*) *Characterizing six seismic solar analogs observed by Kepler, K2, and HERMES*.
9. Ebraheem Farag, Earl Bellinger, et al. (incl. **Christopher Lindsay**), 2026 (*in press, ApJ*) *Self-Consistent Nonlinear Classical Cepheid Pulsations During Stellar Evolution with MESA*.
8. Thiago Ferreira, Earl Bellinger, et al. (incl. **Christopher Lindsay**), 2026 (*in press, ApJ*) *Red-giant Asteroseismology of Low-mass Population III Stars*.
7. Thiago Ferreira, Earl Bellinger, et al. (incl. **Christopher Lindsay**), 2026 (*in press, ApJ*) *Evolution of Low-Mass Population III Stars: Convection, Mass Loss, Nucleosynthesis, and Neutrinos*.
6. Alfio Bonanno, Enrico Cosaro, et al. (incl. **Christopher Lindsay**), 2025 (*in press, ApJ*) *Asteroseismic calibration of the Rossby number and its connection to the stellar dynamo and fundamental properties*.
5. Angharad Weeks, Daniel Huber, et al. (incl. **Christopher Lindsay**), 2025 (*in press, MNRAS*) *Asteroseismology of the ancient naked-eye exoplanet host star ν^2 Lupi*.
4. Claudia Reyes, Dennis Stello, et al. (incl. **Christopher Lindsay**), 2025 (*in press, Nature*) *Acoustic modes in M67 cluster stars trace deepening convective envelopes*.
3. Joel Ong, **Christopher Lindsay**, et al., 2025 (*in press, ApJ*) *Resolving an Asteroseismic Catastrophe: Structural Diagnostics from p-mode Phase Functions off the Main Sequence*.
2. Hans Deeg, et.al. (incl. **C. Lindsay**), 2021 (*in press, JAAVSO*) *Maintaining the Ephemeris of 20 CoRoT Planets: Transit Minimum Times and Potential Transit Timing Variations*
1. J. Cabrera, et.al. (incl. **C. Lindsay**), 2015 (*in press, A&A*) *Transiting exoplanets from the CoRoT space mission XXVII. CoRoT-28b, a planet orbiting an evolved star, and CoRoT-29b, a planet showing an asymmetric transit*

Non-Refereed Papers

1. Y. Asali, K. Gerbig, A. Ghosh, **C. Lindsay**, Z. Shen, and M. Geha, 2022 (*in press, Bulletin of the AAS*, [54\(1\)](#)) *A Standardized Framework for Collecting Graduate Student Input in Faculty Searches*

INVITED TALKS AND SEMINARS

Seminar Talk, The University of Sydney
Stellar Astrophysics and Galactic Archaeology with Asteroseismology

May 1st, 2026
 Sydney, Australia

Seminar Talk, Dartmouth College <i>Stellar Astrophysics and Galactic Archaeology with Asteroseismology</i>	February 2nd, 2026 Hanover, USA
Seminar Talk, University of Alabama <i>Stellar and Galactic Astronomy with Asteroseismology</i>	November 17th, 2025 Tuscaloosa, USA
Thursday Lunch Talk, Princeton University <i>Stellar Astrophysics and Galactic Archaeology with Asteroseismology</i>	October 30th, 2025 Princeton, USA
Seminar Talk, The Ohio State University <i>Stellar Astrophysics and Galactic Archaeology with Asteroseismology</i>	October 3rd, 2025 Columbus, USA
University of Florida Stars and Planets Seminar <i>Stellar Astrophysics and Galactic Archaeology with Asteroseismology</i>	September 30th, 2025 Gainesville, USA
TESS Science Talk at the MIT Kavli Institute <i>Precise Asteroseismic Ages for Red Giants in the Helmi Streams</i>	April 9th, 2025 Cambridge, USA
MIT Monday Afternoon Talk <i>Stellar Astrophysics and Galactic Archaeology with Asteroseismology</i>	April 7th, 2025 Cambridge, USA
Seminar Talk at the American Museum of Natural History <i>Probing the Evolution and Internal Structures of Stars with Asteroseismology</i>	November 14th, 2023 NYC, USA
IfA Astrocoffee Talk at the University of Hawai‘i <i>Probing the Evolution and Internal Structures of Stars with Asteroseismology</i>	January 13th, 2023 Honolulu, USA
TEDx Talk at the University of Southern California <i>Exoplanet Astronomy and the Nobel Prize Hidden in Plain Sight</i>	March 25th, 2022 Los Angeles, USA

FUNDING/PROPOSAL HISTORY

PI, NSF Astronomy & Astrophysics Postdoctoral Fellowship (AAPF) <i>Galactic Archaeology with Precision Asteroseismology</i>	2026-2029 \$330,000
Co-I, TESS General Investigator Program (PI: Joel Ong) <i>Asteroseismic Probes of Convective Boundary Mixing with 200-Second TESS FFIS</i>	2024 \$70,000
Co-I, Yale TAC (PI: W. Garrett Levine) <i>Exoplanet Aeronomy</i>	April 2024 1 Night on Keck/NIRSPEC; 1 Night on Palomar/WIRC
Co-I, TESS General Investigator Program (PI: Joel Ong) <i>Magnetic Activity on Rapidly-Rotating Red Giants with 200-Second TESS FFIS</i>	2023 \$70,000

CONTRIBUTED CONFERENCE PRESENTATIONS

TASC9/KASC16: Contributed Talk <i>Precise Asteroseismic Ages for Red Giants in the Helmi Streams</i>	July 2025 Vienna, Austria
TASC9/KASC16: Poster <i>Asteroseismic Modeling of α-Enhanced, Metal-Poor Stars</i>	July 2025 Vienna, Austria

European Astronomical Society Annual Meeting: Contributed Talk <i>Precise Asteroseismic Ages for the Helmi Stream</i>	June 2025 Cork, Ireland
TESS Science Conference 3 at MIT: Contributed Talk <i>Asteroseismic modeling of metal-poor, α-rich giants in the Halo</i>	July/August 2024 Cambridge, MA
TASC8/KASC15: Contributed Talk <i>Age Dating α-element Enhanced Stars in the Galactic Halo</i>	July 2024 Porto, Portugal
3rd workshop on Artificial Intelligence for Space-Science Research: Virtual Talk May 2024 <i>Detecting Solar Flares in Sun-as-a-Star Observations with CNNs</i>	Melbourne, Australia
TASC7/KASC14: Contributed Talk and Served on the LOC <i>Fossil Signatures of Main-Sequence Convective Cores</i>	July 2023 Honolulu, Hawai'i
TASC6/KASC13: Poster <i>Red Giant Seismology: Seismic Signatures of Convective Overshoot</i>	July 2022 Leuven, Belgium
Cool Stars 21: Poster <i>Mixed Mode Asteroseismology of Red Giants Through the Luminosity Bump</i>	July 2022 Toulouse, France
AAS 240: Contributed Talk <i>Red Giant Seismology: Seismic Signatures of Convective Overshoot</i>	June 15th 2022 Pasadena, CA
TESS Science Conference 2: ePoster <i>Red Giant Seismology</i>	August 2021 Online and Cambridge, MA

ADDITIONAL CONFERENCE AND WORKSHOPS ATTENDED

Porto Summer School on Asteroseismology <i>Attended workshop as a student</i>	July 2024 Via do Conde, Portugal
Life and Death: From Stars to Compact Objects <i>Attended workshop as a student</i>	August-September 2022 Asiago, Italy
MESA Summer School 2022 <i>Attended workshop as a student</i>	August 2022 Santa Barbara, CA
Fifty Years of the Skumanich Relations <i>Attended</i>	February 2022 Boulder, CO

TEACHING EXPERIENCE

ASTR 550: Stellar Astrophysics Teaching Fellow <i>Instructor: Professor Sarbani Basu</i>	Fall 2023, Spring 2025 Yale University
ASTR 120: Galaxies and the Universe Teaching Fellow <i>Instructor: Michael Faison</i>	Spring 2022 Yale University
PHYS 401: Advanced Classical Physics Teaching Fellow <i>Instructor: Professor Nikhil Padmanabhan</i>	Fall 2021 Yale University

ASTR 120: Galaxies & the Universe Teaching Fellow*Instructor: Professor Robert Zinn*

Summer 2021

*Yale University***ASTR 135: Archaeoastronomy Teaching Fellow***Instructor: Michael Faison*

Spring 2021

*Yale University***ASTR 155: Introduction to Astronomical Observing Teaching Fellow***Instructor: Michael Faison*

Fall 2020

*Yale University***ADDITIONAL WORK & SERVICE EXPERIENCE**

Reviewer

2024-Present

- I have reviewed submissions for A&A (3), MNRAS (2), and ApJ (1).
- I also review applications to the Davidson Institute Scholarship.

McDougal Fellow: Graduate Professional Development

August 2024 - May 2026

*Graduate School of Arts and Sciences**Yale University*

- Contributing to the intellectual and professional development of the Yale graduate student community by holding professional development workshops and administering internship opportunities.

Graduate Student Affiliate, Benjamin Franklin College

October 2022 - May 2026

*Benjamin Franklin College**Yale University*

- Promoting interactions between undergraduates and graduate students and working with the Head of College, college staff, and undergraduate students to organize social and educational activities throughout the year.

Chair and Representative, Yale Graduate Student Assembly

May 2021 - May 2024

*Yale Graduate School of Arts & Sciences**Yale University*

- I have served as the Astronomy department representative and Chair of the Yale Graduate School of Arts & Sciences' [Graduate Student Assembly](#).
- I lead the Assembly from May 2023 through May 2024, managing meetings 100+ graduate student representatives and steering our advocacy work in collaboration with the Yale administration.

Local Organizing Committee Member

2023

*Various Conferences**Yale University, University of Hawaii*

- I helped to plan and execute the TASC7/KASC14 asteroseismology workshop at the University of Hawaii in July, 2023.
- I also planned and served as emcee during the Ivy+ Summit of student government leaders at Yale University in October, 2023.

AWARDS, FELLOWSHIPS, AND HONORS

NSF [Astronomy & Astrophysics Postdoctoral Fellowship](#)

September 2026 - Present

Yale [Gruber Science Fellowship](#)

August 2020 - May 2026

USC Order of the Laurel and the Palm	May 2020
USC Provost Research Fellowships	May 2019 - May 2020
USC Summer Undergraduate Research Fund	Summers 2017 & 2018
USC Student Opportunities for Academic Research	August 2016 - May 2018
USC Warren Bennis Scholars Leadership Program	August 2018 - May 2020
USC Trustee Scholar (Full Tuition Remission)	August 2016 - May 2020
Regeneron Science Talent Search (STS) Scholar	2017
Davidson Fellows Laureate (\$50,000 award)	2016